PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2014

Candidate Name	

Civics Group						**INDEX NUMBER**				

H2 BIOLOGY **9648 / 03**

22 Sep 2014 **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.
This document consists of 2 sections:

Section A
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B
Answer Question 5.
Write your answer on the lines provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	**/ 12**
2	**/ 14**
3	**/ 14**
Section B	
5	**/ 20**
TOTAL	**/ 60**
4 (SPA)	**/12**

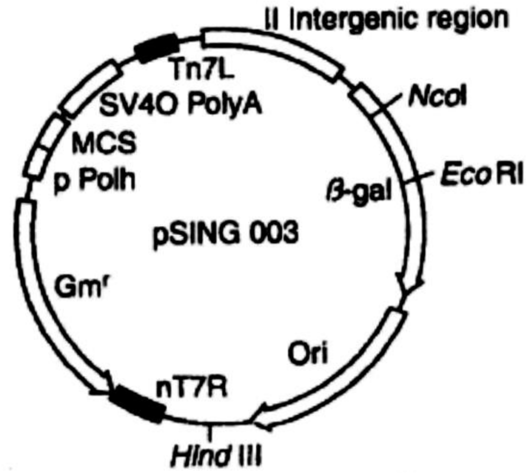
This document consists of **19** printed pages (inclusive of this page)

[Turn Over

Section A: Answer all questions [40 marks]

1. Antithrombin deficiency (AD) is a blood clotting disorder in which sufferers produce insufficient amounts of antithrombin. Antithrombin binds to thrombin and herparin resulting in deactivation of the blood coagulation pathway. AD results in formation of blood clots in blood vessels which obstruct blood flow in the circulatory system.

Fig. 1.1 shows a vector map of the plasmid vector pSING003. The plasmid is used to clone the antithrombin gene, which will be inserted at the *Eco*RI site. The plasmid is subsequently introduced into *E. coli* via transformation. After transformation, the *E. coli* cells are then grown on an agar plate.



Gm^r – gene coding for resistance to the antibiotic gentamycin

β-gal – gene coding for the enzyme β-galactosidase

Ori – origin of replication

Fig. 1.1

(a) Describe the natural function of *Eco*RI restriction enzyme. [1]

A. protect bacteria from attack by <u>viruses/bacteriophages</u> ;
B. cleave any foreign DNA/hydrolyse viral DNA that enters bacterial cell / by cutting up intruding DNA from other organism

(b) In addition to the nutrients required for growth, name **two** other substances that are added to the agar plate to allow selection of cells successfully transformed with the recombinant plasmid. [1]

A. X-gal
B. Antibiotic gentamycin

Fig. 1.2 shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

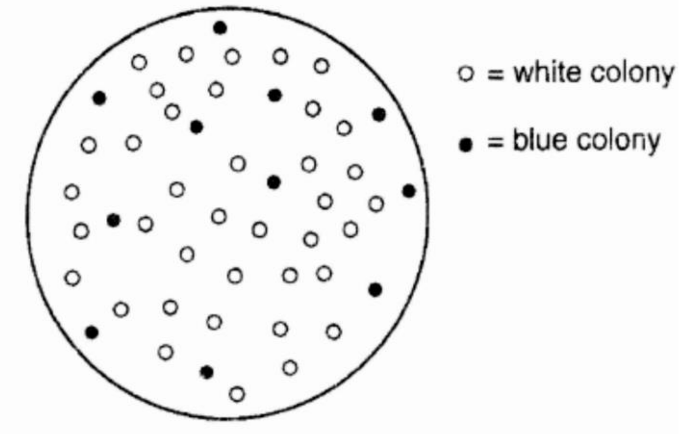


Fig. 1.2

(c) Explain how one would select the successfully transformed colonies. [2]

A. Intact β-galactosidase gene codes for enzyme which hydrolyzes X-gal, giving rise to blue colony (OWTTE);
B. Bacteria colonies with recombinant plasmids will appear white;
C. Due to insertional inactivation which disrupts β-galactosidase gene so;
D. β-galactosidase enzyme is not produced and Xgal is not hydrolysed → white colonies;

RFLP analysis can be used as a method of detecting anti-thrombin deficiency (AD) in family members by using a radioactive probe to determine the presence of the mutant allele that results in AD.

Fig. 1.3 revealed two RFLP morphs that are tightly linked to the AD gene which can be used in the screening of AD. Both RFLP morphs and probe is shown in **Fig. 1.3** with arrows representing the *EcoR1* restriction sites.

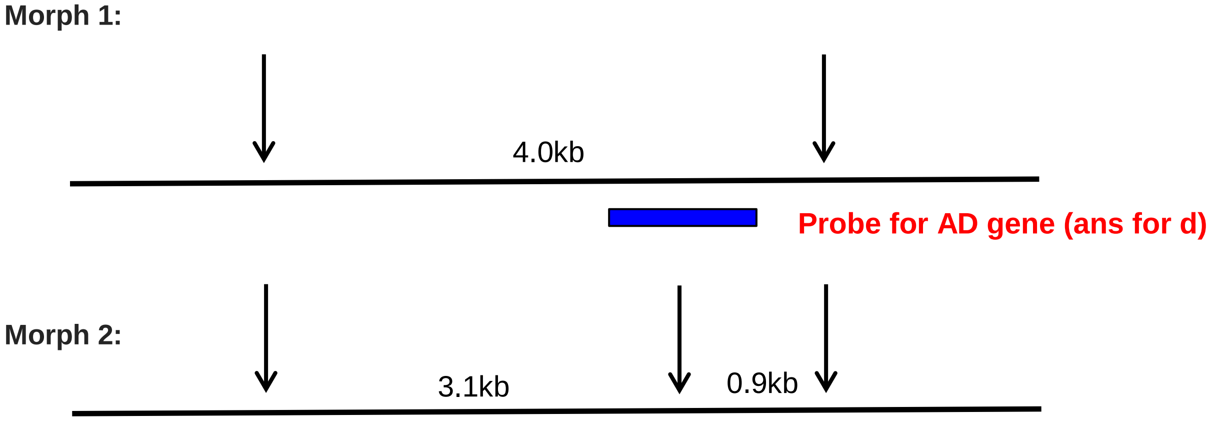


Fig. 1.3

Fig. 1.4 illustrates the Southern Blot analysis of both RFLP morphs in individuals with a family history of AD.

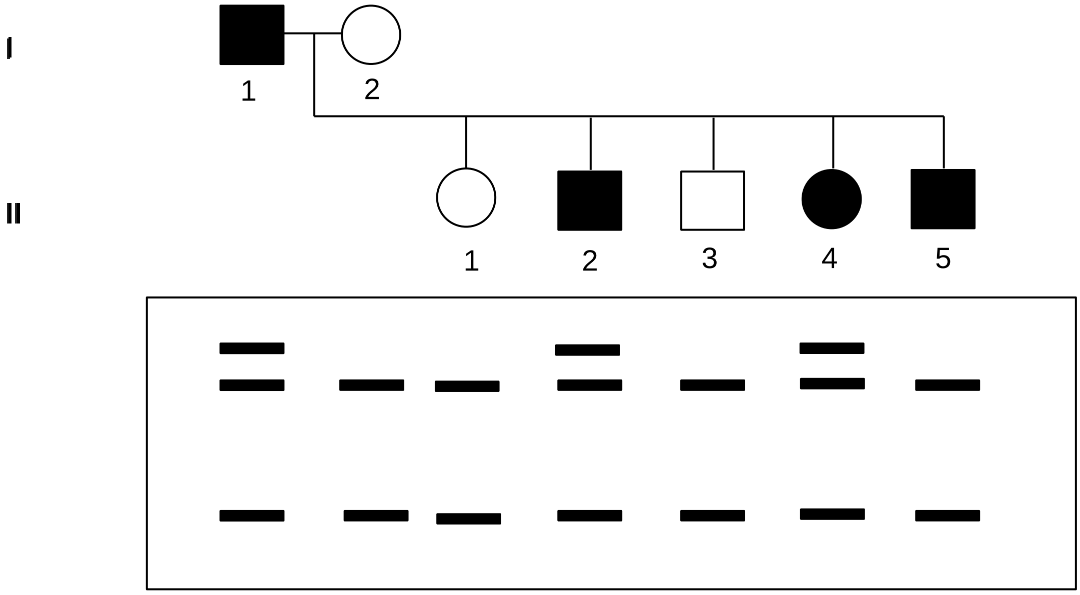


Fig. 1.4

(d) On **Fig. 1.3**, draw the location of the probe that would produce an autoradiogram as indicated in **Fig. 1.4.** [1]

- labeling ;
- drawing of probe at the correct position;

(e) With reference to **Fig. 1.3** and **Fig. 1.4**, state and explain which RFLP morph is linked to the mutant AD gene. [3]

A. RFLP morph 1;
B. AD is an autosomal dominant disease;
C. → this implied that it would require 1 copy of the dominant allele for the individual to inherit the disease;
D. Affected individuals are heterozygous for the disease;
E. Only affected individuals, I1, II2 and II4 have a copy of RFLP morph 1(unaffected individuals do not have 4.0 band) thus a band of 4.0kb is seen and;
F. this band being the largest is the nearest to the well;

(f) Suggest the expected result of the Southern blot analysis of individual II5 and the possible reason for the difference in the DNA patterns for individual II5 from the other affected individuals? [2]

A. a copy of RFLP morph 1 and a copy of morph 2 should be present in individual II5 : thus only morph 1 giving rise to 4.0kb band and morph 2 giving rise to 3.1kb and 0.9kb; OWTTE
B. During gamete formation/meiosis, in one of the parents (I1);
C. Crossing over occurred at loci
D. between RFLP genetic marker and AD mutant gene;
E. This caused AD mutant allele to be linked to RFLP morph 2;

Max 2

(g) State two features of DNA probes that can be used to detect RFLP. [2]

A. DNA probes which are single stranded molecules;;
B. radioactively labeled;;
C. probes complementary in sequence to the RFLP marker;;
(deduct ½ mark if mentioned about complementary to the gene)

@ 1m each, Max 2

[Total: 12 marks]

2. A baby boy was diagnosed with ADA-deficient SCID at age 10 months. The boy had multiple infections, pneumonia, and persistent diarrhea and was not able to gain weight.
He received the enzyme replacement treatment for three to four months, which involved receiving injections twice a week with the necessary enzyme. However, his condition did not improve and he subsequently joined the gene therapy study in 2008. **Fig. 2.1** outlines the gene therapy process.

Baby with mutant alleles of the gene coding for ADA

↓

Stem cells immune system removed from baby

↓ ← Virus carrying normal allele for ADA gene

Some stem cells take up the normal allele

↓

Cells returned to the baby

Fig. 2.1

Today, that boy, who lives with his family in Arizona, is a thriving 5-year-old.

(a) Explain how ADA deficiency results in SCID. [3]

A. Loss-of-function mutation/recessive mutation of gene on chromosome 20;
B. Gene codes for adenosine deaminase, which metabolises dATP;
C. a dysfunctional adenosine deaminase results in the toxic accumulation of dATP in lymphocytes;
D. T cells die and
E. B cells are unable to make antibodies due to the absence of T-cells
F. foreign pathogens cannot be recognized and killed/immune system would be impaired (OWTTE)

(b) Suggest a disadvantage of using the enzyme therapy. [1]

A. Gene in cells is still faulty as this process is not gene therapy – no replacement or introduction of normal functional allele, repeated treatment required OR
B. Enzyme therapy is a life-long process/not a long-term solution – repeated treatment required which can be costly; OR
C. The enzyme may not have entered the cell, in which the dATP was accumulated in - thus ineffective in metabolizing intracellular dATP;

(c) Suggest the appropriate stem cells to be used in this therapy and explain why gene therapy using stem cells was more effective in treating SCID. [2]

A. **Haematopoietic stem cells are undifferentiated;**
B. **These cells can divide/proliferate continuously;**
C. **and can differentiate to form both types of lymphocytes (T & B) carrying normal ADA alleles;**
D. **therefore treatment effects was more long-term;**

(d) Describe how the viral gene delivery system could have been used to insert the normal allele for ADA gene into the stem cells. [3]

RETROVIRUS

A. **A retrovirus could be genetically engineered to remove viral gene which allow for viral replication, assembly and re-infection so that the virus is no longer pathogenic;**
B. **The RNA of the normal functional allele for ADA gene is inserted in place of the viral genes removed (rej: DNA);**
C. **The nucleocapsid also contains reverse transcriptase and integrase to integrate therapeutic gene into genome**
D. **Gene therapy is carried out ex vivo whereby**
E. **Hematopoietic (or blood) stem cells are isolated from the bone marrow of the baby and genetically modified retrovirus is then introduced to these cells**
F. **The virus enters either through fusion;**

ADENOVIRUS

A. **An adenovirus could be genetically engineered to remove viral gene which allow for viral replication, assembly and re-infection so that the virus is no longer pathogenic;**
B. **The DNA of the normal functional allele for ADA gene is inserted in place of the viral genes removed;**
C. **Gene therapy is carried out ex vivo whereby;**
D. **Hematopoietic (or blood) stem cells are isolated from the bone marrow of the baby and genetically modified adenovirus is then introduced to these cells**
E. **The virus enters through receptor-mediated endocytosis;**
F. **The gene exists extrachromosomally;**

(e) Suggest two advantages of using this mode of delivery system as opposed to a non-viral gene delivery system. [2]

A. **Higher efficiency of entering cells/infection;**
B. **since retroviruses/adenovirus are well adapted for infecting cells in the human immune system;**
C. **Can facilitate the movement of normal gene into nucleus followed by the integration of therapeutic genes into the DNA of stem cells (only applicable for retrovirus);**
D. **allowing long-term expression of the ADA gene (only applicable for retrovirus);**

(f) The use of gene therapy to treat diseases has raised many ethical and social issues. Discuss what these issues are. [3]

1. **Social Concerns over gene therapy uses and ramifications leading to health concerns**
   - **It is <u>difficult to determine the long-term effects of exposure to viral vectors</u> and the effects these engineered viruses have on the human genome**
   - **Concerns about many <u>potential dangers and unpredictable factors</u> with gene therapy, which make its practical application risky.**

2. **Discrimination against those who might have the gene but not expressed, → link to social class. (Social)**

3. **Attempts at gene therapy are <u>extremely expensive</u>. (Social)**
   - **This could mean that <u>only the very wealthy and powerful will have access to these therapies due to different social classes.</u>**

4. **The <u>decision of what is normal and what is a disability</u> should not rest on just a small group of individuals or medical professionals. (Ethical)**
   - **The question of whether disabilities are considered diseases and if these need to be cured or prevented require thorough considerations.**
   - **Also, ethical standards for gene therapy may be influenced by politicians, insurance companies and even physicians who may be driven by economical gains.**

5. **<u>Altering one's genes and changing one's genetic makeup goes against many religious and cultural beliefs. (Ethical)</u>**

6. **<u>Exploitation of this technology for aesthetics instead of the original intent of curing a disease (Ethical)</u>**

[Total: 14 marks]

3. Antithrombin factor III is a normal plasma protein (see **Fig 3.1**) that inhibits clotting by binding thrombin and clearing it from the circulation. Persons with born genetic deficiency of antithrombin factor III are at high risk to develop life-threatening clots during high risk situations such as surgery procedures due to thrombosis. Thrombosis is a condition whereby a blood clot forms within a blood vessel, obstructing flow of blood in the circulatory system. Antithrombin factor III is complex structure and often treatment doses are usually large, which posed challenges in collection in the early 1980s.

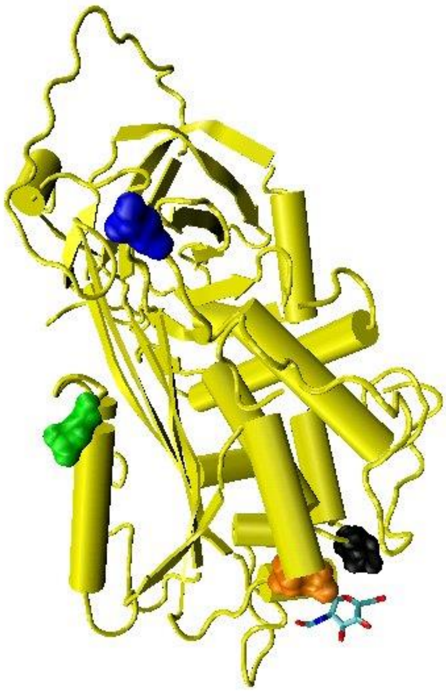


Fig 3.1

Animal pharming is a relatively new technology using whole animals to produce pharmaceutical proteins. A known example is recombinant human antithrombin factor III (rhAT). rhAT presents as an method to manage AT deficiencies by preventing abnormal clotting. A herd of dairy goats were genetically modified to express high levels of human AT in their milk to provide an abundant supply of rhAT.

Beta casein is a protein found in high concentration in goat milk. A schematic representation of the beta casein transgene for recombinant expression of rhAT is shown in **Fig 3.2**. The cDNA encoding rhAT (striped box) replaces part of the coding sequence of beta-casein gene. Restriction sites for enzymes *Eco*RI, *Not*I and *Sal*I are represented by the alphabets R, N and S respectively.

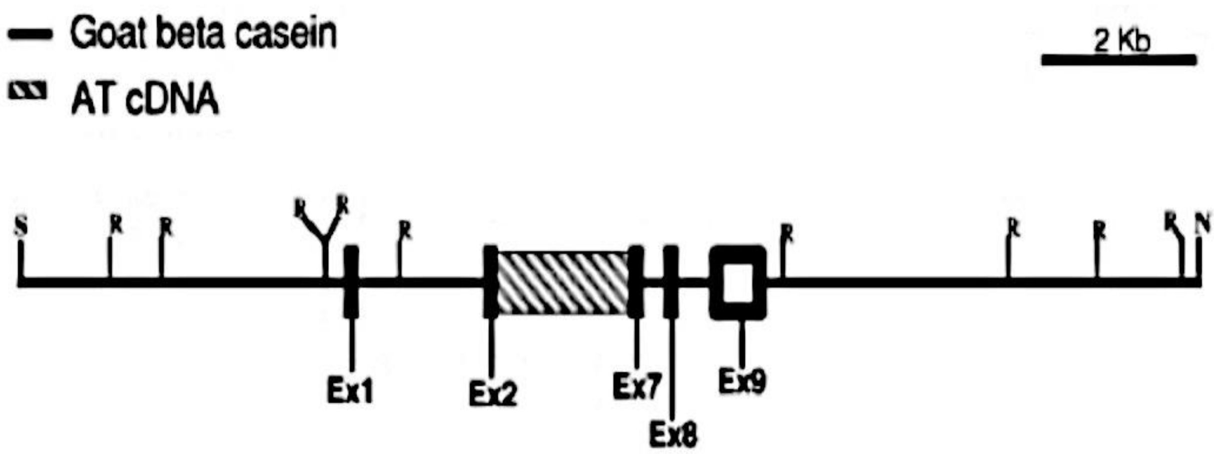


Fig. 3.2

a) Other than restriction sites and genetic markers, what other sequences must be present in this linearized plasmid for it to be an effective expression vector to be inserted into a mammal? Explain your answer. [2]

a. regulatory sequence such as promoter;
b. for the binding of GTF and RNA pol for the assembly of TIC for transcription;
c. Poly-A signal sequence/terminator sequence/3' UTR;
d. For RNA pol to stop transcription/ for mature mRNA to form poly-A tail;

(b) Various animals were considered for the genetic modification as seen in **Table 3.3** below.

Table 3.3

Comparison of the time required to obtain recombinant proteins in different transgenic animal species

	Rabbit	Pig	Sheep	Goat	Cow
Gestation time (months)	1	4	5	5	9
Age at sexual maturity (months)	5	6	8	8	15
Time between gene transfer and first lactation (months)	7	16	18	18	33
Number of offspring	8	10	1–2	1–2	1
Annual milk yield (liters)	15	300	500	800	8000
Recombinant protein per female per year (kg)	0.02	1.5	2.5	4	40

**Gestation refers to the period of developing inside the womb*

(i) What is the advantage of using animal system instead of a bacterial platform? [2]

a. antithrombin factor is a eukaryotic protein;
b. post-transcriptional to excise the introns;
c. post-translational modification is required;
d. to achieve a globular structure;
e. for folding into specific conformation and chemical modification;

Max 2

(ii) Suggest two reasons why goats are used amongst other animals for the production of rhAT. [2]

a. Goat versus sheep,rabbit & pig;
b. goat has a higher yield of 800l of milk as compared to 500l in sheep 300l in pig and 15l in rabbit;
c. Goat versus cow;
d. time between gene transgene and first lactation takes nearly half the time (18 mths vs 33 mths) in goat as compared to cows;
e. Allows pharmaceutical products to be harnessed within a shorter period of time.

Six surgical events on AT-deficient patients were treated with rhAT. The results were shown in **Table 3.4**.

Table 3.4

Table 11.4 Hereditary AT-deficient patients treated with rhAT in a compassionate-use basis

Subject no.	Age [years]	Sex	Baseline AT level [%]	Surgery type	Total rhAT received [U]	Vascular duplex ultra-sound	Clinical evi-dence of thrombosis	Anti rhAT antibody
1a	22	M	40	Laparoscopic splenectomy	74 480	Negative (d5)	Negative	Negative
1b	22	M	40	Bilateral hip replacement	294 000	Not tested	Negative	Negative
2	47	F	49	Hysterectomy	101 921	Negative (d7)	Negative	Negative
3	36	F	58	C-section	65 000	Negative (w6)	Negative	Negative
4	72	M	52	Coronary artery bypass	39 200	Not tested	Negative	Not tested
5	71	M	53	Knee replace-ment	73 480	Negative	Negative	Negative

(c) Comment on the effectiveness of rhAT produced and refined from transgenic goats. [3]

a. rhAT is effective;
b. Successful in preventing thrombosis across all patients;
c. regardless of age, sex and surgery type;
d. It is also safe for usage;
e. as no antibodies are created against the rhAT;
f. indicating that there was no immune response stimulated upon administration of rhAT;

(d) Besides animal pharming, animals have also been genetically modified to increase food yield. A well-known example would be AquAdvantage salmon. AquAdvantage salmon is a transgenic salmon where a promoter for the anti-freeze gene of an ocean pout and a gene for the Chinook growth hormone was inserted into a plasmid pUC18. The recombinant DNA (**Fig. 3.5**) is then introduced into fertilized egg cells of the Atlantic salmon by microinjection and stimulated to become triploid.

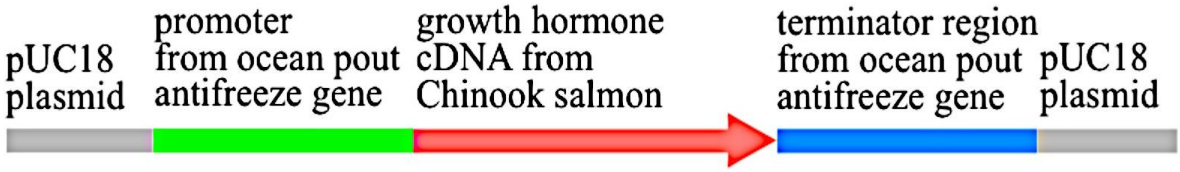


Fig. 3.5

(i) Explain why the promoter from ocean pout antifreeze gene is used. [1]

a. Ocean pout antifreeze gene promoter is a strong promoter;
b. enables growth hormone to be expressed continuously through the year;

(ii) The egg cells were induced to be triploid. Explain the significance of this. [2]

a. AquAdvantage salmon are unable to produce haploid eggs via meiosis;
b. Which renders them infertile;
c. Any AquAdvantage salmon which might escape into the wild will be unable to mate with wild salmon
d. Hence preventing transfer of the transgene to wild species

(iii) Describe one ethical and one social concern that might arise from the production of AquAdvantage salmon. [2]

One each for ethical and social

Health concern (R: Environmental concern as the fish are triploid)
a. The effects of extra salmon growth hormones and the safety of these salmon for human consumption
b. Whether the growth hormones would still be present when consumed by humans.

Ethical concern
a. Increased production of growth hormone has harmful effects on the health of salmon

b. Causing additional stress in the salmon → exploitation of animals for food

[Total: 14 marks]

4. **Planning Question [12 marks]**

A new breed of carrots was tested for sucrose concentration. Freshly squeezed extract was obtained for this investigation.

Your planning must be based on the assumption that you have also been provided with the following materials which you must use:

- Solution of unknown concentration of sucrose extracted from the new breed of carrots, labeled P
- 10% sucrose solution, labeled S
- Distilled water
- Dilute hydrochloric acid
- Sodium hydrogen carbonate powder
- Benedict's solution
- Colorimeter and 1.5 cm^3 cuvettes

A colorimeter is a device that can be used to measure the concentration of a solute by measuring the absorbance of different wavelengths of light in a solution. Generally, a solution of higher concentration will absorb more light energy at a specific wavelength. Therefore, the higher the absorbance value, the higher the concentration of a solution. Although absorbance has no units, it is often recorded as absorbance units (A.U.).

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- syringes
- white tile
- Bunsen burner with tripod, gauze and bench mat
- timer e.g. stopwatch or stop clock
- thermometer

Your plan should

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- an explanation of the theory to support your practical procedure,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record the results and the proposed layout of results tables and graph with clear headings and labels ,
- include reference to safety measures to minimize any risks associated with the proposed experiment,
- use the correct technical and scientific terms.

[Total: 12]

Suggested answers for Planning Question

<u>Aim</u>

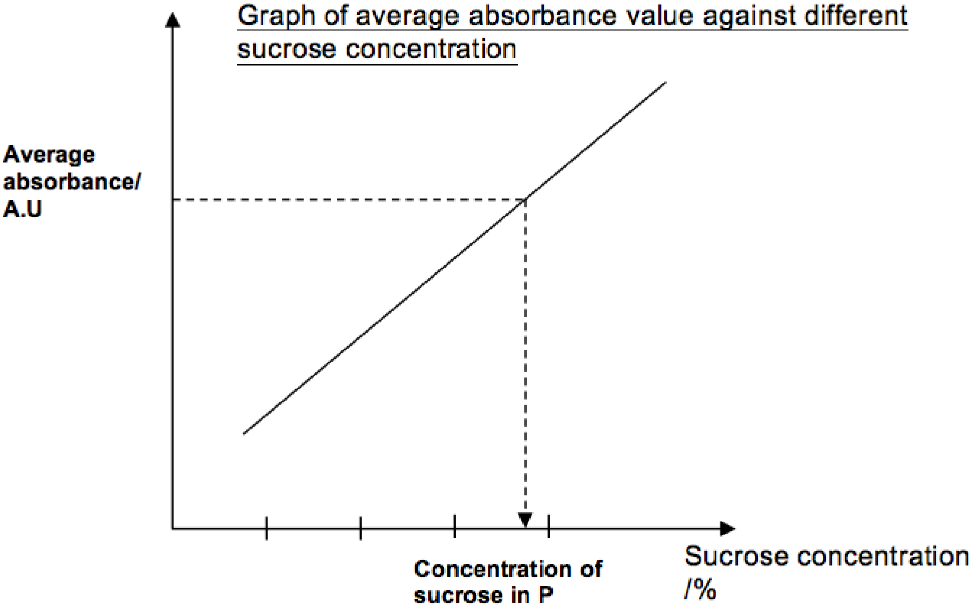
Using colourimeter to measure absorbance, determine the concentration of non-reducing sugar in an extract from a new variety of vegetable.

<u>Background [2.5m, must mention of colorimeter]</u>

a. **Sucrose is a non-reducing sugar disaccharide**
b. **hydrochloric acid with heat can be used to hydrolyse sucrose into glucose and fructose, which are reducing sugars.**
c. **The presence of reducing sugars can be tested using Benedict's solution that contains Cu^{2+}.**
d. **Free carbonyl group in reducing sugar causes Cu^{2+} to be reduced to Cu_2O, which is a red precipitate**
e. **Depending on the amount of glucose and fructose present, varying quantity of red precipitate will be produced as the sugar-benedict solution mixture is immersed in boiling water bath.**
f. **Quantifying the amount of reducing sugars can be achieved through a colorimeter**
g. **Colorimeter is a devise that measures the absorbance of light passing through a solution.**
h. **Solution of higher concentration will absorb more light energy at a specific wavelength.**

<u>Hypothesis [1m]</u>

i. **The higher the concentration of sucrose, the higher the concentration of reducing sugars upon hydrolysis -> greater amount of brick red precipitate -> higher absorbance value**
j. **By plotting a graph of absorbance values of known sucrose concentration**
k. **the concentration of the non-reducing sugar in the extract of the unknown plant can be determined through comparison of absorbance values.**



Procedure [6m]

1. **Prepare the solution in sets of 3**
2. **Prepare 5 different sucrose solutions of known concentration using simple dilution/serial dilution, refer to the table below:**

Sucrose concentration/%	Volume of 10% stock sucrose/cm^3	Volume of distilled water/cm^3
0	0	10
2	2	8
4	4	6
6	6	4
8	8	2
10	10	0

Interval between concentrations should be equal

3. **Use syringe to add 2cm^3 of each sucrose solution (including solution S) into 5 separate test tubes.**
4. **Label these test tubes in accordance to the concentration of sucrose solution that they contain.**
5. **Use syringe to add 1cm^3 dilute hydrochloric acid to each sucrose solution and mix well.**
6. **Place the test tubes into a boiling water bath for 1 minute, immediately start timing using a stop watch.**
7. **Remove the test tubes from water bath and allow the tubes to cool on a test tube rack.**
8. **Use spatula to add small amount of solid sodium hydrogen carbonate until no effervescence is observed. This serves to neutralize the hydrochloric acid.**
9. **Use syringe to add 5cm^3 Benedict solution to each test tube.**

10. Place test tubes into a boiling water bath for 3 minutes. Place test tube on a test-tube rack after removal from the water bath.
11. Calibrate the colorimeter using a blank curvette filled with distilled water.
12. Place $2cm^3$ of solution from step 10 into the cuvette and read the absorbance value using a colorimeter at fixed wavelength. Ensure homogeneity of the solution before extracting $2cm^3$ with a pipette.
13. Repeat step 11 for all concentrations of sucrose.
14. Repeat steps 2 to 11 twice for a replicate readings in sets of three (0.5 mark)
15. Obtain a standard curve with the averages of the readings
16. Repeat the experiment from step 3 to 11 three times with $2cm^3$ of extract from the unknown plant
17. Compare the average absorbance value against the standard curve
18. Repeat step 1 to 13 at least twice for reproducibility of results, using fresh stock solution S. (0.5 mark)

Results

Sucrose concentration /%	Absorbance value / a.u.			
	Repeat 1	Repeat 2	Repeat 3	Average
0				
2				
4				
6				
8				
10				
P (unknown)				

*2 tables must be present (1/2 mark each)

Variables [1.5M]

- Independent Variable: sucrose concentration/% (0,2,4,6,8,10)
- Dependent Variable: absorbance value of amount of reducing sugar produced /A.U
- Controlled variables (can be mentioned in procedure also):
  - volume of sucrose solution used
  - volume of HCl used and duration of hydrolysis reaction
  - volume of Benedict's solution added and duration of boiling
  - Temperature of hydrolysis reaction and Benedict's Test

Control [0.5m]

- Control: to test if P contains reducing sugar that will give color with Benedict's test→ conduct a Benedict's test with P directly without acid hydrolysis. Should there be color, this absorbance reading must be deducted from the absorbance after hydrolysis.

Risk Assessment [0.5m]

- **Do not allow reagents such as Benedict's solution, dilute hydrochloric acid to have contact with skin/eyes as these can cause skin irritation. Wash when in contact**
- **Boiling water bath can scald. Handle hot glassware such as test tubes with test-tube holders or oven gloves/rags.**
- **Handle hot glassware such as test tubes and beakers appropriately, e.g. use test tubes holder or gloves as hot glassware can cause burns and hot liquid can cause scalding.**
- **Light Bunsen burner with care; turn off when not in used.**

Max 12 marks

Section C: Essay Question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate.
- must be set out in section (a), (b), etc., as indicated in the question.

5(a) Outline, using a named example, how the micropropagation technique is used in production and cloning of pest resistant plants. [6]

1. **Obtain mRNA for BT gene and using reverse transcription method, produce the double stranded cDNA**
   - Add linker molecules to both ends of the gene of interest
2. **Remove the oncogenes and opine producing genes from the Ti plasmids of Agrobacterium.**
3. **cut the Ti plasmid using the same restriction enzymes as that for Bt gene, forming complementary sticky ends**
4. **Mix the cut plasmid and gene of interest to allow complementary base-pairing via hydrogen bonds to temporarily hold the fragments together**
5. **Add DNA ligase to seal the nicks in the sugar-phosphate backbone forms phosphodiester linkages between fragments**
6. **Insert recombinant Ti-plasmids back into the Agrobacterium**
7. **In micropropagation, plant tissues or explants eg: meristematic cells are removed from plants ® explants only;**
8. **Infect the explants with with Agrobacterium containing the recombinant Ti plasmids. Select explants that were successfully transformed;**
9. **Surface of explants sterilised with dilute sodium hypochlorite (Clorox) to kill bacterial and fungal pathogens or organisms;**
10. **And grown in a containing sterile media containing nutrients and plant growth regulators needed for plant growth;**
11. **Containers holding the explants are then sealed and incubated for 1-9 weeks.**
12. **During this period, the cultured cells divide by mitosis**

13. to form a mass of undifferentiated tissue called a callus.
14. As callus increases in size, pieces of callus is sliced off and grown on new medium composition (subculturing);
15. By adjusting concentration of auxin:cytokinin ratio in growth medium,
16. cells in callus can be induced to differentiate into roots and shoots;
17. Further growth being encouraged by the use of plant growth substances until plantlets are large enough
18. To be weaned and planted in sterile soil in green house

    After acclimatization in green house, the plant are transferred to soil for field planting

(b) Describe how RFLP analysis is used to support the process of linkage mapping. [6]

1. By linkage analysis using RFLP as genetic markers (which have known physical location on a chromosome);
2. Construct map using recombination frequency to determine order/sequence of genes and relative distance between different genes and genetic markers along a chromosome;

Basis and role of Linkage Mapping:

3. A linkage map shows the relative location;
4. or the order of genes along a chromosome;
5. constructed on the assumption that the probability of a crossover between two genetic loci/RFLP markers is proportional to the distance separating the loci;
6. Linkage maps are usually constructed with several thousand known genetic markers spaced evenly throughout the genome.
7. The RFLP markers can be any genes or any other identifiable DNA sequences, such as variation number tandem repeats (VNTR) and short tandem repeats (STR).

How to carry out linkage mapping using RFLP:

8. Linkage map must be done based on experimental crosses;
9. After mating/fertilization, RFLP from parental and offspring organisms are obtained and analysed;
10. Using gel E & Southern blot/nucleic acid hybridisation;
11. RFLP pattern obtained are used to calculate the total percentage of recombinant offspring;
12. Give an indication of the distance between the RFLP markers/sequences based on the recombination frequencies obtained;
    (a) the farther apart the two RFLP markers/sequences are, the higher the probability that a crossover that generates genetic variation will occur between them and therefore
    (b) the higher the recombination frequency; & *vice versa*
    (c) For example, if 70% of the progeny produced are parental and 30% were recombinant, the two RFLP loci are 30 centi-Morgans (cM) apart from each other

(c) Describe, using a named example, how RFLP enables disease detection. [8]

1. Sickle cell anaemia

2. **Individuals who have the disease will carry a specific allele of an RFLP, suggesting that the RFLP marker/locus is linked to the disease gene locus for haemoglobin.**
3. **Genome from a normal individual, carrier (heterozygous) and affected individual is digested with the same restriction enzyme, MstII.**
4. **Differences in DNA sequence of both samples results in differences in restriction enzyme recognition site and result in restriction fragments of different lengths.**
5. **MstII restriction site (CCTNAGG) or CTT**
6. **Sickle cell allele – 1.3kbp fragment due to loss of restriction site. A is substituted by T, changing the sequence from CCTGAGG to CCTGTGG, thus the loss of a restriction site. GAG → GTG ; CTT→ CAT**
7. **Normal allele – presence of one restriction site, two fragments – 1.1kbp and 0.2 kbp.**
8. **DNA restriction fragments are separated according to size / molecular weight and analysed by agarose gel electrophoresis & nucleic acid hybridisation.**
   (a) **Smaller fragments migrate faster and vice versa**
   (b) **Agarose gel is placed in alkaline solution to separate DNA into single-stranded DNA and single-stranded DNA is transferred onto nitrocellulose membrane**
   (c) **Blot is incubated with single-stranded radioactively labeled nucleic acid probes which will hybridize to specific target nucleotide sequences by complementary base pairing**
   (d) **Autoradiography is carried out and X-ray film is exposed to X-ray. Target DNA bands bound to radioactively labelled probes will show up as a black band on the X-ray film.**
9. **A carrier would be heterozygous for the condition (one normal and one mutated allele on each homologous chromosome), and thus three bands would be observed on the gel. (State the sizes of fragments -1.3kb, 1.1kb and 0.2kb)**
10. **A normal individual would be homozygous for the normal allele, hence only two bands would show up on the gel, corresponding to the sizes of 1.1kbp and 0.2kpb in the DNA ladder.**
11. **An affected individual, homozygous for the mutated allele, would have only one band, nearest to the well, corresponding to 1.3kbp fragment.**

It is difficult to come up with the same detailed answer for cystic fibrosis as there seems to be little information on the restriction enzyme, mutation and other details of rflp analysis for cystic fibrosis.

The End